

Problem #1: Radiation Transport (18 points)

The most important form of energy transport for stars on the main sequence (which is 90% of the stellar lifetime), is radiation transport.

Problem #1a: How old are those photons? (8 points)

Even though we don't always get to see them through Pittsburgh's clouds, photons are leaving the surface of the Sun all the time. Estimate the minimum time it takes a photon to travel from the core of the Sun to the surface, assuming that the photon retains its identity every time it is absorbed and re-emitted. This is probably not realistic, but I haven't yet had the opportunity to ask a photon if it remembers all the places it has been absorbed and emitted.

(i) First, calculate the mean free path of a photon in the interior regions of the Sun noting that the minimum opacity of fully ionized hydrogen is set by the electron scattering cross section to be $\kappa_{\text{es}} = 0.4 \text{ cm}^2/\text{g}$. *Hint:* You can use the average density that you calculated in Homework 3 for a $1 M_{\odot}$ star. (2 points)

(ii) Next, using the speed of the photon, the mean free path, and the radius of the sun, derive an equation for the time it takes for the photon to travel from the center of the sun to its surface. I think the easiest way to do this is using the dimensions of each variable, but you might think a different way is easiest! (4 points)

(iii) How does your result compare to the thermal timescale of the Sun? (2 points)

Problem #1b: Conceptual stuff (4 points)

Why does the diffusion approximation for radiation transport break down near the surface of a star? Supply your answer in full at least one full sentence.

Problem #1c: How small is a 'small' temperature gradient? (6 points)

Estimate the temperature gradient, $\partial T/\partial r$, in a gas blob inside of the Sun where $T \sim 10^7 \text{ K}$, $r \sim 0.2 R_{\odot}$, $l(r) \sim 3.8 \times 10^{33} \text{ erg/s}$, $\kappa_{\text{es}} \sim 0.4 \text{ cm}^2/\text{g}$. What is the change in the temperature over the range of a single mean free path of a photon?

Problem #2: Figures and captions (20 points)

Your project presentations are coming up soon! As part of your presentation, you should plan to include at least one original figure that you (and your partner if you have one) make. In the presentation, you'll be able to walk the audience through the figure. This isn't the only way that figures are presented though!

For this problem, you should create a figure (you and your partner can submit the same figure) that you could use in your presentation. The figure should illustrate some conclusion or result from your project and should have adequate axis labels and a legend if necessary. Once you have a figure, you should also write a figure caption that describes the necessary information for a reader to understand and interpret the figure themselves. It's also a good idea to add in a punchline that you want your reader to take away from the figure. *You must write your own figure caption even if you and your partner share the figure.*

Hint! If you need some inspiration, take a look at the some figures in the papers you searched for on ads in homework 2.